Pranav Sakhuja

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Education

University of California, San Diego San Diego, CA
*Bachelor of Science, **Mechanical Engineering with a specialization in Controls and Robotics*** *Awarded June 2025*
GPA: 3.76/4.00

Carnegie Mellon University Pittsburgh, PA
*Master of Science, **Mechanical Engineering - Research Track** (Incoming Student)* *Starting Fall 2026*

Skills

Programming: Python, MATLAB, Arduino C, Bash, Linux Systems, Robot Operating System (ROS and ROS2), SQL
Software Tools: Simulink, SOLIDWORKS, Autodesk Fusion, AutoCAD
Manufacturing: FDM/SLA 3D Printing, CNC and Manual Machining, Laser Welding (TPU), CAM, DFM, GD&T
Coursework & Technical Knowledge: Control Theory, Path Planning & Estimation, Machine Learning, Project Management

Work Experience

Morimoto Lab, UC San Diego San Diego, CA
Research Assistant *May 2023 – Present*

- Developing a minimally invasive optical-biopsy endoscope using a soft, tip-growing vine-robot body, integrating a millimeter-scale camera and rotating polarizer setup for colonoscopic applications
- Developed a novel vacuum-driven, mechanically programmable actuator to enhance steering performance of vine robots
- Procured and configured the da Vinci Research Kit (dVRK) hardware and ROS-based software for surgical robotics research
- Built a ROS-based teleoperation simulation integrating Concentric Tube Robots (CTRs) with the dVRK's manipulators to improve dexterity and expand laparoscopic surgery workspaces

KPIT Technologies Inc. Novi, MI
Mechanical Engineering Intern — Vehicle Engineering & Design Team *June 2024 – September 2024*

- Conducted in-depth benchmarking studies of Thermal Management System (TMS) architectures for various Battery Electric Vehicles (BEVs)
- Developed advanced plant models using MATLAB, Simulink, and Simscape, enhancing simulation accuracy and enabling comprehensive analysis of BEV-TMS performance
- Optimized various control architectures and schemes, resulting in improved thermal efficiency and system responsiveness for BEV applications under harsh weather conditions by integrating TMS models with powertrain performance models

Triton NeuroTech at UC San Diego San Diego, CA
President *November 2022 – June 2025*

- Directed the management, technical advising, and timeline execution for 12+ research-focused Neuroengineering, Brain-Computer Interface (BCI) projects, involving over 200 members
- Conducted technical workshops on data acquisition, signal processing, hardware architecture design, enabling the development of biosensing devices using EEG/EMG signals
- Led the organizing team for the 2025 California Neurotechnology Conference, hosting 330+ attendees and featuring high-impact speakers, researchers, and leaders from academia and industry

Projects

Soft Robotic Optical Biopsy Endoscope *Morimoto Lab, UC San Diego*

- Optimized manufacturing process to fabricate a 1.5m long, 12mm diameter vine robot endoscope body with 2 retraction channels using a novel laser welding technique for Thermoplastic Polyurethane (TPU)
- Designing and implementing a camera module, incorporating a dexterous rotating polarizer mechanism for minimally invasive cancerous tissue detection

Autonomous Autogyro UAV *MAE Senior Capstone, UC San Diego*

- Led design and implementation of wiring harness, flight controls, autopilot integration, and Ground Control System for an autonomous autogyro UAV capable of real-time surveillance, navigation, control, and computer vision
- Optimized payload integration and flight dynamics to support onboard camera, telemetry, and data acquisition modules

Autonomous Robotic Car *MAE 148, UC San Diego*

- Integrated NVIDIA Jetson, camera, LiDAR, and image-perception neural networks for vehicle navigation and obstacle avoidance using ROS 2
- Implemented Visual SLAM for real-time perception and object search in unknown indoor environments

Neural Prosthetic Robotic Arm *Presented at California Neurotechnology Conference 2023*

- Designed and developed an EMG-controlled prosthetic arm using SOLIDWORKS, followed by 3D printing and assembly
- Led mechatronic system design by programming Arduinos, signal processing in Python, soldering circuitry, implementing serial communication for motor control with NVIDIA Jetson-hosted ML pipeline